

A COMPLETE DEPTH-TWO ATLAS FOR FIRST-DESCENT PREFIX REVERSAL

ANONYMOUS

ABSTRACT. Given a nonidentity permutation, locate its first descent and reverse the prefix ending at the descent's smaller follower. We determine the complete functional graph of this autonomous map. Its image is the half of $\mathfrak{S}_n$ beginning with an ascent. For $n \geq 3$, the recurrent set is the identity together with the $n!/3$ permutations having a peak at position two; the latter form $n!/6$ two-cycles. Every tail has length at most two, with exact populations $n!/3 + 1, n!/2, n!/6 - 1$. The full predecessor set of a target is indexed by the decreasing run starting at its second position. Consequently, for $n \geq 4$ the maximum fibre is $n - 1$, attained at exactly $n - 1$ targets. The exceptional $n = 2, 3$ atlases are explicit. Arbitrary prefix reversal, pancake sorting, descent counts, and finite-map bookkeeping are treated as background. The residual conjunction is OWNER_AMBER, and external circulation is HOLD_EXTERNAL.

1. THE AUTONOMOUS RULE AND THEOREM PACKAGE

For $1 \leq k \leq n$, let ρ_k reverse the first k entries of a permutation:

$$(1) \quad \rho_k(\pi_1 \cdots \pi_k \pi_{k+1} \cdots \pi_n) = \pi_k \cdots \pi_1 \pi_{k+1} \cdots \pi_n.$$

Write $\iota = 12 \cdots n$. If $\pi \neq \iota$, let

$$(2) \quad d(\pi) = \min\{i : \pi_i > \pi_{i+1}\}$$

and define the first-descent prefix-reversal map

$$(3) \quad F_n(\iota) = \iota, \quad F_n(\pi) = \rho_{d(\pi)+1}(\pi) \quad (\pi \neq \iota).$$

The extra position after the first descent is part of the rule.

Prefix reversal and its pancake-sorting distance are classical [1]; prefix-reversal sorting on fixed-alphabet strings is also established [2]. Pudwell and Smith study four shuffle maps selected by a cut after the longest increasing prefix [3], but those maps do not reverse the full prefix in (3). We assign arbitrary prefix reversal, pancake graphs and distances, longest-increasing-prefix selection, elementary descent and peak counts, and generic finite-map bookkeeping no contribution credit.

There is also a sharp negative control. The rule called “First Sort” in Project Euler Problems 523–524 moves the follower π_{d+1} to the front; it does not reverse the selected prefix.¹ For example, it sends 1324 to 2134, whereas (3) sends 1324 to 2314. We therefore do not use “First Sort” as a name for F_n .

For $n \geq 3$, set

$$(4) \quad \mathcal{I}_n = \{\tau \in \mathfrak{S}_n : \tau_1 < \tau_2\}, \quad \mathcal{P}_n = \{\tau \in \mathfrak{S}_n : \tau_1 < \tau_2 > \tau_3\}, \quad \mathcal{R}_n = \{\iota\} \cup \mathcal{P}_n.$$

Let $\mu(\pi)$ be the distance from π to the recurrent set of F_n . For $\tau \in \mathcal{I}_n$, let

$$(5) \quad r(\tau) = \max\{r : 1 \leq r \leq n - 1, \tau_2 > \tau_3 > \cdots > \tau_{r+1}\},$$

where the condition is empty when $r = 1$.

Theorem 1 (complete functional-graph atlas). *For every $n \geq 3$, the following hold.*

- (i) *The image is exactly $\mathcal{I}_n$, and $|\mathcal{I}_n| = n!/2$.*
- (ii) *The recurrent set is exactly $\mathcal{R}_n$. The identity is fixed, while the $n!/3$ states in $\mathcal{P}_n$ form $n!/6$ disjoint two-cycles under ρ_3 .*
- (iii) *Every tail has length at most two, and*

$$(6) \quad \#\{\mu = 0\} = \frac{n!}{3} + 1, \quad \#\{\mu = 1\} = \frac{n!}{2}, \quad \#\{\mu = 2\} = \frac{n!}{6} - 1.$$

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¹Official statements: <https://projecteuler.net/problem=523> and <https://projecteuler.net/problem=524>.

(iv) A target outside $\mathcal{I}_n$ has no predecessor. For $\tau \in \mathcal{I}_n$, the complete set of distinct predecessors is

$$(7) \quad F_n^{-1}(\tau) = \{\rho_k(\tau) : 2 \leq k \leq r(\tau) + 1\} \cup \begin{cases} \{\iota\}, & \tau = \iota, \\ \emptyset, & \tau \neq \iota. \end{cases}$$

Consequently,

$$(8) \quad |F_n^{-1}(\tau)| = r(\tau) + \mathbf{1}_{\{\tau=\iota\}}.$$

(v) For $n \geq 4$, the maximum fibre size is $n - 1$. Its complete maximizing set is

$$(9) \quad \mathcal{M}_n = \{\tau \in \mathfrak{S}_n : \tau_2 = n, \tau_2 > \tau_3 > \cdots > \tau_n\}, \quad |\mathcal{M}_n| = n - 1.$$

For $n = 3$, the maximum remains $2 = n - 1$, but the identity is one additional maximizer beyond the two states in (9).

The five conclusions can be read at a glance as follows.

object	exact description
image	$\tau_1 < \tau_2$; population $n!/2$
recurrent core	$\{\iota\} \cup \{\tau_1 < \tau_2 > \tau_3\}$; one fixed point and $n!/6$ two-cycles
tail populations	$n!/3 + 1, n!/2, n!/6 - 1$ at depths 0, 1, 2
target fibre	decreasing-run reversals in (7)
maximum, $n \geq 4$	size $n - 1$ at the $n - 1$ targets in (9)

2. HALF-IMAGE AND RECURRENT CORE

Proof of Theorem 1(i). If $d = d(\pi)$, the first two entries of $F_n(\pi)$ are $\pi_{d+1} < \pi_d$. Thus every nonfixed output begins with an ascent; so does ι . Hence $\text{im } F_n \subseteq \mathcal{I}_n$.

Conversely, take $\tau \in \mathcal{I}_n$ and put $\pi = \rho_2(\tau)$. The first comparison of π is the descent $\tau_2 > \tau_1$, so $F_n(\pi) = \rho_2(\pi) = \tau$. This proves equality. Swapping the first two entries is a fixed-point-free involution pairing permutations in $\mathcal{I}_n$ with those beginning with a descent, and therefore $|\mathcal{I}_n| = n!/2$. $\square$

Proof of Theorem 1(ii). For $\tau \in \mathcal{P}_n$, the first descent is at position two. Hence $F_n(\tau) = \rho_3(\tau)$. The new first three entries are $\tau_3 < \tau_2 > \tau_1$, so the image is again in $\mathcal{P}_n$; applying ρ_3 once more returns to τ . The two states are distinct because their first and third entries cannot agree.

Now let $\tau \in \mathcal{I}_n \setminus \mathcal{R}_n$. Since it is neither a peak at two nor the identity, its first three entries increase and its first descent occurs at some $d \geq 3$. The first three entries of $F_n(\tau)$ are

$$(10) \quad \tau_{d+1} < \tau_d > \tau_{d-1},$$

so $F_n(\tau) \in \mathcal{P}_n$. Every state first enters $\mathcal{I}_n$ by part (i), and every image state then lies in or enters $\mathcal{R}_n$. This proves that no other cycle exists.

Among the six relative orders of the first three entries, exactly two put their maximum in the middle. Each order occurs for $n!/6$ permutations, so $|\mathcal{P}_n| = n!/3$. Pairing by ρ_3 gives $n!/6$ two-cycles. $\square$

3. TARGET FIBRES AND EXACT TAIL CENSUS

Lemma 2 (decreasing-run inverse). *For $\tau \in \mathcal{I}_n$ and $2 \leq k \leq n$, the state $\rho_k(\tau)$ has first descent at position $k - 1$ if and only if $k \leq r(\tau) + 1$. These valid states are distinct as k varies.*

Proof. The first k entries of $\rho_k(\tau)$ are

$$(11) \quad \tau_k, \tau_{k-1}, \dots, \tau_2, \tau_1.$$

They increase through position $k - 1$ precisely when $\tau_2 > \tau_3 > \cdots > \tau_k$. The next comparison is then the descent $\tau_2 > \tau_1$, which holds because $\tau \in \mathcal{I}_n$. This condition is equivalent to $k \leq r(\tau) + 1$. Different k give different first entries τ_k , proving distinctness. $\square$

Proof of Theorem 1(iv). Part (i) excludes predecessors for targets outside $\mathcal{I}_n$. Suppose $\tau \in \mathcal{I}_n$. Every nonfixed predecessor π has some first descent d and satisfies $\tau = \rho_{d+1}(\pi)$; since prefix reversal is an involution, $\pi = \rho_{d+1}(\tau)$. Lemma 2 gives exactly the values $2 \leq d + 1 \leq r(\tau) + 1$, and it also proves the converse for every such value. The only fixed state is ι , which contributes one further predecessor exactly when $\tau = \iota$. For that target, $\rho_2(\iota)$ is distinct from ι , so (8) follows. $\square$

Proof of Theorem 1(iii). Part (ii) gives the depth-zero population $n!/3 + 1$ and shows that every image state outside $\mathcal{R}_n$ has depth one. Such a state τ begins $\tau_1 < \tau_2 < \tau_3$, so $r(\tau) = 1$. Lemma 2 says that its only predecessor is $\rho_2(\tau)$, which begins with a descent and lies outside $\mathcal{I}_n$. The map

$$(12) \quad \mathcal{I}_n \setminus \mathcal{R}_n \longrightarrow \{\pi : \mu(\pi) = 2\}, \quad \tau \longmapsto \rho_2(\tau)$$

is therefore a bijection. Indeed, every depth-two state must map to a nonrecurrent image state, so the displayed construction also captures the reverse inclusion.

Using parts (i)–(ii), the depth-two population is

$$(13) \quad |\mathcal{I}_n| - |\mathcal{R}_n| = \frac{n!}{2} - \frac{n!}{3} - 1 = \frac{n!}{6} - 1.$$

All remaining states have depth one, giving $n! - (n!/3 + 1) - (n!/6 - 1) = n!/2$. This proves (6) and the absence of longer tails. $\square$

4. SHARP FIBRES AND THE SMALL BOUNDARIES

Proof of Theorem 1(v). For a nonidentity image target, (8) is at most $n - 1$; the identity has fibre two. When $n \geq 4$, equality can therefore occur only when $r(\tau) = n - 1$, or equivalently $\tau_2 > \tau_3 > \dots > \tau_n$. Since also $\tau_1 < \tau_2$, the value n cannot occupy the first position. It lies in the decreasing tail and must be its first, largest entry: $\tau_2 = n$. Conversely, every target in (9) has the full run and hence fibre $n - 1$. Choose τ_1 arbitrarily from $[n - 1]$; the remaining entries are then forced in decreasing order. This gives exactly $n - 1$ maximizers.

At $n = 3$, the two full-run targets still have fibre two, and the identity's extra fixed predecessor also gives fibre two. This is the stated exception. $\square$

Proposition 3 (the $n = 2, 3$ atlases). *At $n = 2$, the arrows are $12 \mapsto 12$ and $21 \mapsto 12$. Thus the image and recurrent set are both $\{12\}$, the depth populations are $(1, 1)$, and $F_2^{-1}(12) = \{12, 21\}$ is the unique maximum fibre.*

At $n = 3$, the full functional graph is

$$(14) \quad 123 \mapsto 123, \quad 132 \leftrightarrow 231, \quad 213 \mapsto 123, \quad 312 \mapsto 132, \quad 321 \mapsto 231.$$

The image and recurrent populations are both three, the depth populations are $(3, 3, 0)$, and each image target has fibre two. Thus the three maximizers are 123, 132, 231.

Proof. Both displays follow by applying (3). Their inverse lists also agree with (7): the identity has its fixed predecessor in addition to its two-prefix reversal. This direct calculation isolates the only boundary not covered by the $n \geq 3$ recurrent count and the only tie not covered uniformly by the $n \geq 4$ maximizer statement. $\square$

5. EXACT CONTROLS AND CLAIM BOUNDARY

The paper-local verifier reconstructs the full map, all incoming sets, and all orbit coordinates through $\mathfrak{S}_9$. It makes 6,273,063 integer assertions, including complete checks of the five theorem contracts and the $n = 2, 3$ boundaries. It imports no scouting or earlier-paper code. These finite boxes are falsification pressure, not proofs or ownership evidence.

The nearest internal proof vocabulary is P122's deterministic permutation block reversal and target-local inverse cuts; that vocabulary receives zero separation credit. The killed follower-to-front spike is the Project Euler rule, not (3). The retained owner-thin conjunction is the literal autonomous reversal together with its half-image, peak two-cycle core, depth census, and decreasing-run inverse atlas. A bounded search non-hit is not a novelty result. The gate remains OWNER_AMBER, and the external lifecycle remains HOLD_EXTERNAL.

REFERENCES

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